THE RIGHT DERIVED FUNCTOR OF THE PUSHFORWARD AGREES WITH THE DERIVED PUSHFORWARD ON THE DERIVED CATEGORY OF QUASI-COHERENT MODULES

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Allowed standard prerequisites (editorial).

- For a quasi-compact scheme with affine diagonal the derived category of quasi-coherent modules agrees with the full derived category of quasi-coherent sheaves. invoked at proof lines 1842-1846. Lets the proof replace one category by the other and compare the two functors on the same objects.
- A pushforward along an affine morphism whose source category has an exact left adjoint preserves K-injective complexes, so it can be used to compute derived pushforwards. invoked at proof lines 1849-1862. Shows that pushing forward a K-injective complex of quasi-coherent modules from an affine open is again K-injective and represents its derived pushforward.
- The pushforward along an affine morphism of a quasi-coherent complex represents its derived pushforward. invoked at proof lines 1858-1864. Identifies the values of both functors on the complexes coming from an affine open.

Source reference. Original proposition 08DB: sources/stacks-perfect/pinned-original-source.tex, source lines 1801–1801. Cited at proof lines 1842-1845 for the equivalence of the categories; the comparison of the functors once they are identified is inside the counted span.

Source reference. Original lemma 01SG: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 1856-1858 to see that the inclusion of an affine open and its composite with the given morphism are affine; the K-injectivity argument built on that is inside the counted span.

Source reference. Original lemma 08BJ: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 1859-1862 to conclude that the pushforward from the affine open preserves K-injective complexes; the computation that follows is inside the counted span.

Source reference. Original lemma 02MZ: sources/stacks-perfect/pinned-original-source.tex, source lines 1580–1580. Cited at proof lines 1863-1866 to identify the

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two functors on complexes pushed forward from the affine open; the generation argument extending the conclusion to all objects is inside the counted span.

Lemma 0.2. *Let $f : X \to Y$ be a morphism of schemes. Assume X and Y are quasi-compact and have affine diagonal. Then, denoting*

$$\Phi : D(QCoh(\mathcal{O}_X)) \to D(QCoh(\mathcal{O}_Y))$$

the right derived functor of $f_* : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_Y)$ the diagram

$$\begin{array}{ccc} D(QCoh(\mathcal{O}_X)) & \longrightarrow & D_{QCoh}(\mathcal{O}_X) \\ \downarrow{\scriptstyle \Phi} & & \downarrow{\scriptstyle Rf_*} \\ D(QCoh(\mathcal{O}_Y)) & \longrightarrow & D_{QCoh}(\mathcal{O}_Y) \end{array}$$

is commutative.

Proof. Observe that the horizontal arrows in the diagram are equivalences of categories by Proposition 08DB. Hence we can identify these categories (and similarly for other quasi-compact schemes with affine diagonal). The statement of the lemma is that the canonical map $\Phi(K) \to Rf_*(K)$ is an isomorphism for all K in $D(QCoh(\mathcal{O}_X))$. Note that if $K_1 \to K_2 \to K_3 \to K_1[1]$ is a distinguished triangle in $D(QCoh(\mathcal{O}_X))$ and the statement is true for two-out-of-three, then it is true for the third.

Let $U \subset X$ be an affine open. Since the diagonal of X is affine, the inclusion morphism $j : U \to X$ is affine (Morphisms, Lemma 01SG). Similarly, the composition $g = f \circ j : U \to Y$ is affine. Let $\mathcal{I}^\bullet$ be a K-injective complex in $QCoh(\mathcal{O}_U)$. Since $j_* : QCoh(\mathcal{O}_U) \to QCoh(\mathcal{O}_X)$ has an exact left adjoint $j^* : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_U)$ we see that $j_*\mathcal{I}^\bullet$ is a K-injective complex in $QCoh(\mathcal{O}_X)$, see Derived Categories, Lemma 08BJ. It follows that

$$\Phi(j_*\mathcal{I}^\bullet) = f_*j_*\mathcal{I}^\bullet = g_*\mathcal{I}^\bullet$$

By Lemma 02MZ we see that $j_*\mathcal{I}^\bullet$ represents $Rj_*\mathcal{I}^\bullet$ and $g_*\mathcal{I}^\bullet$ represents $Rg_*\mathcal{I}^\bullet$. On the other hand, we have $Rf_* \circ Rj_* = Rg_*$. Hence $f_*j_*\mathcal{I}^\bullet$ represents $Rf_*(j_*\mathcal{I}^\bullet)$. We conclude that the lemma is true for any complex of the form $j_*\mathcal{G}^\bullet$ with $\mathcal{G}^\bullet$ a complex of quasi-coherent modules on U. (Note that if $\mathcal{G}^\bullet \to \mathcal{I}^\bullet$ is a quasi-isomorphism, then $j_*\mathcal{G}^\bullet \to j_*\mathcal{I}^\bullet$ is a quasi-isomorphism as well since j_* is an exact functor on quasi-coherent modules.)

Let $\mathcal{F}^\bullet$ be a complex of quasi-coherent $\mathcal{O}_X$-modules. Let $T \subset X$ be a closed subset such that the support of $\mathcal{F}^p$ is contained in T for all p. We will use induction on the minimal number n of affine opens $U_1, \ldots, U_n$ such that $T \subset U_1 \cup \ldots \cup U_n$. The base case $n = 0$ is trivial. If $n \geq 1$, then set $U = U_1$ and denote $j : U \to X$ the open immersion as above. We consider the map of complexes $c : \mathcal{F}^\bullet \to j_*j^*\mathcal{F}^\bullet$. We obtain two short exact sequences of complexes:

$$0 \to \mathrm{Ker}(c) \to \mathcal{F}^\bullet \to \mathrm{Im}(c) \to 0$$

and

$$0 \to \mathrm{Im}(c) \to j_*j^*\mathcal{F}^\bullet \to \mathrm{Coker}(c) \to 0$$

The complexes $\mathrm{Ker}(c)$ and $\mathrm{Coker}(c)$ are supported on $T \setminus U \subset U_2 \cup \ldots \cup U_n$ and the result holds for them by induction. The result holds for $j_*j^*\mathcal{F}^\bullet$ by the discussion

in the preceding paragraph. We conclude by looking at the distinguished triangles associated to the short exact sequences and using the initial remark of the proof. □